

Experiment No.	6
Experiment Title	Bagging, Boosting, and Stacked Ensemble Models
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1 Objective

To understand the theoretical foundations and operational mechanics of primary ensemble learning paradigms: **Bagging** (Bootstrap Aggregation), **Boosting** (Adaptive Boosting and Gradient Tree Boosting), and **Stacking** (Stacked Generalization).

- Implement a Bagging Classifier with Decision Tree base estimators and analyze the influence of bootstrap parameters.
- Implement and evaluate sequential Boosting Classifiers (AdaBoost and Gradient Boosting) across tuned learning rates and depths.
- Construct a heterogeneous Stacked Ensemble Model integrating Support Vector Machines (SVM), Gaussian Naive Bayes, and Decision Trees via Logistic Regression meta-learner.
- Benchmark ensemble architectures against individual base classifiers using 5-fold Stratified Cross-Validation and bias-variance decomposition.

2 Problem Statement

To classify breast tumors as either *Malignant* or *Benign* based on cytological attributes extracted from Fine Needle Aspirate (FNA) digitized microscopic biopsy images, comparing how Bagging, Boosting, and Stacking improve predictive reliability, generalization capacity, and stability compared to single baseline learners.

The public **Wisconsin Diagnostic Breast Cancer (WDBC)** dataset comprising 569 clinical biopsy records characterized by 30 continuous morphometric features is used for this experiment.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)
Dataset Source	UCI Machine Learning Repository / Scikit-Learn
Number of Samples	569 patient biopsy cases
Number of Features	30 continuous cytological descriptors
Target Classes	2 (Malignant: 212 [37.3%], Benign: 357 [62.7%])
Missing Values	0 missing records
Machine Learning Task	Supervised Binary Classification
Train-Test Split	80% Training (455 samples), 20% Testing (114 samples)

4 Exploratory Data Analysis

Inference: Target distribution comprises 357 benign and 212 malignant records. Stratified sampling preserves this 62.7% to 37.3% ratio across cross-validation folds.

Inference: Severe multicollinearity exists among nuclear size descriptors ($r > 0.90$ between radius, perimeter, and area). Tree ensembles robustly navigate collinearity via random feature subspaces.

5 Hyperparameter Tuning Results

Table 1: Bagging Classifier Hyperparameter Tuning

Base Estimator	Estimators (M)	Max Samples	Max Features	Validation F1
Decision Tree (max_depth=None)	10	0.8	0.8	0.9571
Decision Tree (max_depth=None)	50	0.8	0.8	0.9682
Decision Tree (max_depth=None)	100	1.0	1.0	0.9726
Decision Tree (max_depth=None)	200	0.8	1.0	0.9789

Table 2: Boosting Classifiers Hyperparameter Tuning

Model	Estimators (M)	Learning Rate (eta)	Max Depth	Validation F1
AdaBoost	50	0.1	1 (Stump)	0.9655
AdaBoost	100	0.5	1 (Stump)	0.9722
AdaBoost	200	1.0	1 (Stump)	0.9790
Gradient Boosting	50	0.05	3	0.9650
Gradient Boosting	100	0.1	3	0.9724
Gradient Boosting	200	0.1	3	0.9861

Table 3: Stacking Classifier Architecture Configuration

Layer	Constituent Estimators	Tuned Hyperparameters
Level-0 Base Learners	SVM (SVC(probability=True))	C=1.0, kernel='rbf', gamma='scale'
Level-0 Base Learners	Gaussian Naive Bayes	var_smoothing=1e-9
Level-0 Base Learners	Decision Tree	max_depth=5, criterion='entropy'
Level-1 Meta-Learner	Logistic Regression	C=1.0, penalty='l2', solver='lbfgs'

6 Comprehensive Model Benchmark (Table 4)

Model Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Baseline Decision Tree	0.9386	0.9452	0.9583	0.9517	0.9317
Support Vector Machine (RBF)	0.9737	0.9726	0.9861	0.9793	0.9957
Gaussian Naive Bayes	0.9386	0.9333	0.9722	0.9524	0.9888
Bagging Classifier	0.9649	0.9722	0.9722	0.9722	0.9934
AdaBoost Classifier	0.9737	0.9726	0.9861	0.9793	0.9924
Gradient Tree Boosting	0.9649	0.9595	0.9861	0.9726	0.9947
Stacked Ensemble Model	0.9825	0.9861	0.9861	0.9861	0.9974

7 Performance Visualizations

8 Discussion

- **Observation:** Ensemble architectures consistently outperform single baseline decision trees. The stacked classifier achieved the highest accuracy (0.9825) and ROC-AUC (0.9974).
- **Strength:** Bagging effectively controls estimator variance, while Boosting sequentially minimizes residual bias. Stacking leverages diverse hypotheses from SVM, Naive Bayes, and Decision Trees.
- **Limitation:** Boosting is more sensitive to outliers and label noise than Bagging. Stacking requires multi-fold cross-validation generation of out-of-fold meta-features.
- **Improvement:** Computational performance can be enhanced with early stopping in gradient boosting and automated threshold tuning for clinical cost-sensitive classification.

9 Experimental Setup

Python Version	3.11.x
Scikit-Learn Version	1.6.1
IDE	Google Colab / Local Jupyter Environment
Operating System	Windows 11
Hardware Platform	x86_64 Multicore Processor

10 References

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